Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

At a certain temperature 2 moles of carbon monoxide and 3 moles of chlorine were allowed to reach equilibrium according to the reaction $CO + Cl_2 \rightleftharpoons COCl_2$ in a 5 lit vessel. A equilibrium if one mole of CO is present then equilibrium constant for the reaction is:

(A) 2

(B) 2.5

(C) 3.0

(D) 4

Q.2

Which of the following reactions represent a heterogenous equilibrium?

(A) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

(B) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

(C) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

(D) $2O_3(g) \rightleftharpoons 3O_2(g)$

Q.3

For a reaction if $K_p > K_c$, the forward reaction is favoured by

(A) Low pressure

(B) High pressure

(C) High temperature

(D) Low temperature

Q.4

Naphthalene, a white solid used to make mothballs, has a vapour pressure of $0.10 mmHg$ at $27^\circ C$. Hence, and K_p and K_c for the equilibrium are : $C_{10}H_8(s) \rightleftharpoons C_{10}H_8(g)$

(A) $0.10, 0.10$

(B) $0.10, 4.1 \times 10^{-3}$

(C) $1.32 \times 10^{-4}, 5.34 \times 10^{-6}$

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(D) $5.36 \times 10^{-6}, 1.32 \times 10^{-4}$

Q.5

According to law of mass action rate of a chemical reaction is proportional to

(A) Concentration of reactants

(B) Molar concentration of reactants

(C) Concentration of products

(D) Molar concentration of products

Q.6

$\mathrm{K_c}$ for the reaction $\mathrm{N_2(\ g) + O_2(\ g) \rightleftharpoons 2NO(g)}$
at $300\ \mathrm{K}$ is 4.0×10^{-6}.$\mathrm{K_p}$ for the above reaction will be $\left(\mathrm{R} = 2\mathrm{calmol^{-1}\ K^{-1}}\right)$

(A) 2.4×10^{-3}

(B) 4×10^{-6}

(C) $4 \times 10^{-6}(\mathrm{RT})^2$

(D) 16×10^{-12}

Q.7

The following reaction is performed at $298\ \mathrm{K}$. $\mathrm{2NO(g) + O_2(\ g) \rightleftharpoons 2NO_2(\ g)}$
The standard free energy of formation of $\mathrm{NO(g)}$ is $86.6\ \mathrm{kJ/mol}$ at $298\ \mathrm{K}$. What is the standard free energy of formation of $\mathrm{NO_2(\ g)}$ at $298\ \mathrm{K}$? $\left(K_p = 1.6 \times 10^{12}\right)$

(A) $86600 - \frac{\ln\left(1.6 \times 10^{12}\right)}{\mathrm{R}(298)}$

(B) $0.5\left[2 \times 86,600 - \mathrm{R}(298) \ln\left(1.6 \times 10^{12}\right)\right]$

(C) $\mathrm{R}(298) \ln\left(1.6 \times 10^{12}\right) - 86600$

(D) $86600 + \mathrm{R}(298) \ln\left(1.6 \times 10^{12}\right)$

Q.8

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The equilibrium constant of the following are :

$$N_2 + 3H_2 \rightleftharpoons 2NH_3 \qquad K_1$$
$$N_2 + O_2 \rightleftharpoons 2NO \qquad K_2$$
$$H_2 + \frac{1}{2}O_2 \rightarrow H_2O \qquad K_3$$

The equilibrium constant (K) of the reaction :
$2NH_3 + \frac{5}{2}O_2 \overset{K}{\rightleftharpoons} 2NO + 3H_2O$, will be

(A) $K_2\ K_3^3/K_1$

(B) $K_2\ K_3/K_1$

(C) $K_2^3\ K_3/K_1$

(D) $K_1\ K_3^3/K_2$

Q.9

The degree of dissociation of $PCl_5(\alpha)$ obeying the equilibrium $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ is related to the equilibrium pressure by

(A) $\alpha \propto \frac{1}{P^4}$

(B) $\alpha \propto \frac{1}{\sqrt{P}}$

(C) $\alpha \propto \frac{1}{P^2}$

(D) $\alpha \propto P$

Q.10

A 20 litre container at $400\ K$ contains $CO_2(g)$ at pressure $0.4\ atm$ and an excess of SrO (neglect the volume of solid SrO). The volume of the container is now decreased by moving the movable piston fitted in the container. The maximum volume of the container, when pressure of CO_2 attains its maximum value, will be :-
(Given that : $SrCO_3(s) \rightleftharpoons SrO(s) + CO_2(g), Kp = 1.6\ atm$)

(A) 10 litre

(B) 4 litre

(C) 2 litre

(D) 5 litre

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Q.11

For the reaction $\mathrm{H_2(\ g) + I_2(\ g) \rightleftharpoons 2HI(g)}$ at $\mathrm{721\ K}$, the value of equilibrium constant is 50 , when equilibrium concentration of both is $\mathrm{5M}$. Value of $\mathrm{K_p}$ under the same conditions will be

(A) 0.02

(B) 0.2

(C) 50

(D) $\mathrm{50RT}$

Q.12

For a reversible gaseous reaction $\mathrm{N_2 + 3H_2 \rightleftharpoons 2NH_3}$ at equilibrium, if some moles of $\mathrm{H_2}$ are replaced by same number of moles of $\mathrm{T_2}$ (T is tritium, isotope of H and assume isotopes do not have different chemical properties) without affecting other parameter, then:

(A) the sample of ammonia obtained after sometime will be radioactive.

(B) moles of $\mathrm{N_2}$ after the change will be different as compared to moles of $\mathrm{N_2}$ present before the change

(C) the value of $\mathrm{K_p}$ or $\mathrm{K_c}$ will change

(D) the average molecular mass of new equilibrium will be same as that of old equilibrium

Q.13

The ratio $\frac{K_p}{K_c}$ for the reaction $\mathrm{CO}(g) + \frac{1}{2}\mathrm{O_2}(g) \rightleftharpoons \mathrm{CO_2}(g)$ is:

(A) $\frac{1}{\sqrt{RT}}$

(B) $(RT)^{1/2}$

(C) RT

(D) 1

Q.14

Which of the following is true at chemical equilibrium?

(A) $(\Delta \mathrm{G})_{\mathrm{T,P}}$ is minimum and $(\Delta \mathrm{S})_{\mathrm{U,V}}$ is also minimum

(B) $(\Delta \mathrm{G})_{\mathrm{T,V}}$ is minimum and $(\Delta \mathrm{S})_{\mathrm{U,V}}$ is maximum

(C) $(\Delta \mathrm{G})_{\mathrm{T,V}}$ is maximum and $(\Delta \mathrm{S})_{\mathrm{U,V}}$ is zero

(D) $(\Delta \mathrm{G})_{\mathrm{T,P}}$ is zero and $(\Delta \mathrm{S})_{\mathrm{U,V}}$ is also zero

Q.15

K_1, K_2 and K_3 are the equilibrium constants of the following reactions (I), (II) and (III) respectively:

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(I) $\mathrm{N_2 + 2O_2 \rightleftharpoons 2NO_2}$
(II) $\mathrm{2NO_2 \rightleftharpoons N_2 + 2O_2}$
(III) $\mathrm{NO_2 \rightleftharpoons \frac{1}{2}\, N_2 + O_2}$
The correct relation from the following is

(A) $K_1 = \frac{1}{K_2} = \frac{1}{K_3}$

(B) $K_1 = \frac{1}{K_2} = \frac{1}{(K_3)^2}$

(C) $K_1 = \sqrt{K_2} = K_3$

(D) $K_1 = \frac{1}{K_2} = K_3$

Q.16

Consider the expression $\mathrm{\Delta G = -RTlnK_p + RTlnQ_p}$ and indicate the correct statement at equilibrium

(A) $\mathrm{\Delta G = 0, Q_p > K_p}$ the equilibrium reaction will shift from left to right

(B) $\mathrm{\Delta G = 0, Q_p = K_p}$ the equilibrium reaction will shift from left to right

(C) $\mathrm{\Delta G = \infty, Q_p < K_p}$ the equilibrium reaction will shift from right to left

(D) $\mathrm{\Delta G < 0, Q_p > K_p}$ the equilibrium reaction will shift from right to left where Q_p and K_p term refer to reaction quotient and equilibrium constant at constant pressure respectively.

Q.17

If the synthesis of ammonia from Haber's process is carried out with exactly the same starting conditions (of partial pressure and temperature) but using $\mathrm{D_2}$ (deuterium) in place of $\mathrm{H_2}$. Then

(A) the equilibrium will be disturbed

(B) the composition of reaction mixture will remain same at equilibrium.

(C) Use of isotope in reaction will not produce ammonia.

(D) At equilibrium rate of forward reaction will be greater than the rate of reverse reaction

Q.18

$8\ \mathrm{mol}$ of $AB_3(g)$ are introduced into a $1.0\mathrm{dm}^3$ vessel. If it dissociates as $2AB_3(g) \rightleftharpoons A_2(g) + 3B_2(g)$. At equilibrium, 2 mol of A_2 are found to be present. The equilibrium constant of this reaction is

(A) 2

(B) 3

(C) 27

(D) 36

Q.19

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One mole of $O_2(g)$ and two moles of $SO_2(g)$ were heated in a closed vessel of one-litre capacity at $1098\ K$. At equilibrium 1.6 moles of $SO_3(g)$ were found. The equilibrium constant K_c of the reaction would be

(A) 30

(B) 40

(C) 80

(D) 60

Q.20

The increase of pressure on ice $\rightleftharpoons$ water system at constant temperature will lead to

(A) a decrease in the entropy of the system

(B) an increase in the Gibb's energy of the system

(C) no effect on the equilibrium

(D) a shift of the equilibrium in the forward direction

Answers & Solutions

Q.1 Answer:

2.5

Solution:

	CO	$+$	$Cl_2 \rightleftharpoons$	$COCl_2$
At t = 0	2		3	0
At equilibrium	(2–1)		(3–1)	1
Concentrations (V = 5 lit)	$\frac{1}{5}$		$\frac{2}{5}$	$\frac{1}{5}$

$K_c = \frac{[COCl_2]}{[CO][Cl_2]} = \frac{1/5}{\frac{1}{5}\times\frac{2}{5}} = \frac{25}{10}$ **or** $K_c = 2.5$

Q.2 Answer:

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

Solution:

In heterogenous equilibrium, physical state of all the reactants and products are not same

Q.3 Answer:

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Low pressure

Solution:

No solution available

Q.4 Answer:

$1.32 \times 10^{-4}, 5.34 \times 10^{-6}$

Solution:

$\mathrm{K_p} = \mathrm{P_{C_{10}H_8(\,g)}}$
$= 0.10\ \mathrm{mm} = \frac{0.10}{760}\ \mathrm{atm} = 1.32 \times 10^{-4}$
$\mathrm{K_c} = \frac{\mathrm{K_p}}{(\mathrm{RT})^{\mathrm{Ln}}} = \frac{\mathrm{K_p}}{\mathrm{RT}} = \frac{1.32 \times 10^{-4}}{0.0821 \times 300}$
$= 5.36 \times 10^{-6}$

Q.5 Answer:

Molar concentration of reactants

Solution:

According to law of mass-action, "at a given temperature, the rate of a reaction at a particular instant is proportional to the product of the active masses of the reactants at that instant raised to powers which are numerically equal to the numbers of their respective molecules in the stoichiometric equation describing the reaction".

Q.6 Answer:

4×10^{-6}

Solution:

$\mathrm{K_p} = \mathrm{K_c}(\mathrm{RT})^{\Delta \mathrm{n}}$ **since** $\Delta \mathrm{n} = 0,\ \mathrm{K_p} = \mathrm{K_c}$.

Q.7 Answer:

$0.5\left[2 \times 86,600 - \mathrm{R}(298) \ln\left(1.6 \times 10^{12}\right)\right]$

Solution:

$\Delta \mathrm{G}^{\circ}_{\mathrm{NOg}} = 86.6\ \mathrm{kJ/mol} = 86600\ \mathrm{J/mol}$
$\mathrm{G}^{\circ}_{\mathrm{NO_2(\,g)}} = \mathrm{xJ/mol}$
$\mathrm{T} = 298,\ \mathrm{K_p} = 1.6 \times 10^{12}$
$\Delta \mathrm{G}^{\circ} = -\mathrm{RT} \ln \mathrm{K_P}$
Given equation, $2\mathrm{NO(g)} + \mathrm{O_2(\,g)} \rightleftharpoons 2\mathrm{NO_2(\,g)}$
$\therefore \quad 2\Delta \mathrm{G}^{\circ}_{\mathrm{NO_2}} - 2\Delta \mathrm{G}^{\circ}_{\mathrm{NO}} = -\mathrm{R}(298) \ln\left(1.6 \times 10^{12}\right)$
$2\Delta \mathrm{G}^{\mathrm{NO_2}}_{\mathrm{NO_2}} - 2 \times 86600 = -\mathrm{R}(298) \ln\left(1.6 \times 10^{12}\right)$
$2\Delta \mathrm{G}^{\mathrm{OO_2}}_{\mathrm{NO_2}} = 2 \times 86600 - \mathrm{R}(298) \ln\left(1.6 \times 10^{12}\right)$

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$\Delta G^{\circ}_{NO_2} = \frac{1}{2}\left[2 \times 86600 - R(298)\ln\left(1.6 \times 10^{12}\right]\right.$
$= 0.5\left[2 \times 86600 - R(298)\ln\left(1.6 \times 10^{12}\right)\right]$

Q.8 Answer:

$K_2\ K_3^3/K_1$

Solution:

$N_2 + 3H_2 \rightleftharpoons 2NH_3; K_1 = \frac{[NH_3]^2}{[N_2][H_2]^3}$
(ii) $N_2 + O_2 \rightleftharpoons 2NO; K_2 = \frac{[NO]^2}{[N_2][O_2]}$
(iii) $H_2 + \frac{1}{2}O_2 \longrightarrow H_2O; K_3 = \frac{[H_2O]}{[H_2][O_2]^{1/2}}$
Applying (II $+3\times$ III $-$ I) we will get $2NH_3 + \frac{5}{2}O_2 \overset{K}{\rightleftarrows} 2NO + 3H_2O$
$K = \frac{[NO]^2}{[N_2][O_2]} \times \frac{[H_2O]^3}{[H_2]^3 \times [O_2]^{3/2}} / \frac{[NH_3]^2}{[N_2][H_2]^3}$
$\therefore K = K_2 \times K_3^3/K_1$

Q.9 Answer:

$\alpha \propto \frac{1}{\sqrt{P}}$

Solution:

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$
$1-\alpha \quad \alpha \quad \alpha$
$\therefore K_p = \frac{\frac{\alpha}{1+\alpha}P \times \frac{\alpha}{1+\alpha}P}{\frac{1-\alpha}{1+\alpha}P} = \frac{\alpha^2 P}{1-\alpha^2}$
or, $K_p = \alpha^2 P \quad \therefore \alpha = \sqrt{\frac{K_p}{P}}$ **when** $1 - \alpha^2 = 1$

Q.10 Answer:

5 litre

Solution:

Max. pressure of CO_2 **= Pressure of** CO_2 **at equilibrium For reaction,** $SrCO_3(s) \rightleftharpoons SrO(s) + CO_2$
$K_p = P_{CO2} = 1.6\ atm =$ **maximum pressure of** CO_2
volume of container at this stage. $V = \frac{nRT}{P}$ **...(i)**
Since container is sealed and reaction was not earlier at equilibrium. $\therefore n =$ constant
$n = \frac{PV}{RT} = \frac{0.4 \times 20}{RT}$**...(ii)**
Put equation (ii) in equation (i) $V = \left[\frac{0.4 \times 20}{RT}\right]\frac{RT}{1.6} = 5\ L$

Q.11 Answer:

50

Solution:

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$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
$K_p = K_c(RT)^{\Delta n}$;
$\Delta n = 2 - 2 = 0; \therefore K_p = K_c$

Q.12 Answer:

the sample of ammonia obtained after sometime will be radioactive.

Solution:

Since T_2 has similar chemical properties as H_2 so upon mixing, $NH_2\ T$ and NHT_2 are formed.

Q.13 Answer:

$\frac{1}{\sqrt{RT}}$

Solution:

$K_P = K_C(RT)^{\Delta n_g}$

For the reaction

$CO(g) + \frac{1}{2}O_2(g) \leftrightharpoons CO_2(g)$
$\Delta n_g = 1 - \left(1 + \frac{1}{2}\right) = -\frac{1}{2}$
$\therefore \quad K_P = \frac{K_C}{\sqrt{RT}}; \quad \frac{K_P}{K_C} = \frac{1}{\sqrt{RT}}$

Q.14 Answer:

$(\Delta G)_{T,P}$ is zero and $(\Delta S)_{U,V}$ is also zero

Solution:

The condition for the equilibrium is

$(\Delta G)_{T,P} = 0$ **and** $(\Delta S)_{U,V} = 0$

Q.15 Answer:

$K_1 = \frac{1}{K_2} = \frac{1}{(K_3)^2}$

Solution:

$N_2 + 2O_2 \overset{K_1}{\rightleftharpoons} 2NO_2$
$K_1 = \frac{[NO_2]^2}{[N_2][O_2]^2}$ **...(i)**
(II) $2NO_2 \overset{K_2}{\rightleftharpoons} N_2 + 2O_2$
$K_2 = \frac{[N_2][O_2]^2}{[NO_2]^2}$ **...(ii)**
(III) $NO_2 \overset{K_3}{\rightleftharpoons} \frac{1}{2}N_2 + O_2$
$K_3 = \frac{[N_2]^{1/2}[O_2]}{[NO_2]}$
$\therefore (K_3)^2 = \frac{[N_2][O_2]^2}{[NO_2]^2}$ **...(iii)**
from equation (i), (ii) and (iii) $K_1 = \frac{1}{K_2} = \frac{1}{(K_3)^2}$

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Q.16 Answer:

$\Delta G = 0, Q_p = K_p$ the equilibrium reaction will shift from left to right

Solution:

Van't Hoff reaction isotherm is $\Delta G = \Delta G^\circ + RT \ln Qp$

When the reaction is in a state of equilibrium $\Delta G = 0$

Then $\Delta G^o = -RT \ln Q_p = -RT \ln K_p$

Q.17 Answer:

the composition of reaction mixture will remain same at equilibrium.

Solution:

The reaction mixtures starting either with H_2 **or** D_2 **reach equilibrium with the same composition, except that** D_2 **and** ND_3 **are present instead of** H_2 **and** NH_3**.**

Q.18 Answer:

27

Solution:

$$2AB_3(g) \rightleftharpoons A_2(g) + 3B_2(g)$$

at $t = 0$	8	0	0
at eq.	$(8 - 2 \times 2)$	2	3×2
	$= 4$	2	6

now

$K_C = \frac{[A_2][B_2]^3}{[AB_3]^2} = \frac{2/1 \times [6/1]^3}{[4/1]^2} = 27$

Q.19 Answer:

80

Solution:

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	O_2	+ $2SO_2$	$\rightleftharpoons$ $2SO_3$
at t = 0	1	2	0
at equil.	$(1-\alpha)$	$2(1-\alpha)$	2α

Given at equilibrium $2\alpha = 1.6$

$\alpha = 0.8$

$K_C = \frac{(2\alpha)^2}{(1-\alpha)(2-2\alpha)^2} = \frac{(0.8)^2}{(1-0.8)(1-0.8)^2}$

$= \frac{0.64}{0.008}$

$K_C = 80$

Q.20 Answer:

a shift of the equilibrium in the forward direction

Solution:

Volume of ice is greater than that of water. The direction in which the reaction will proceed can be predicted by applying Le-Chatelier's principle Pressure $\propto \frac{1}{\text{Volume}}$ **So equilibrium, will shift forward.**

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